

A Suppression-based STDP Rule Resilient to Jitter Noise in Spike Patterns for Neuromorphic Computing

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Abstract—Multi-spike models of synaptic plasticity, such as the triplet and suppression spike-timing-dependent plasticity (STDP) rules, exhibit better alignment with neurophysiological data in the brain compared to the pair-based STDP rule. Previous studies have empirically shown that the pair-based STDP rule can detect spatiotemporal spike patterns hidden in equally dense distractor spike trains in an unsupervised manner. However, it fails to detect spike patterns influenced by jitter noise. Given that spiking neural networks (SNNs) exhibit variability in generated spike trains in response to the same inputs, it becomes imperative to have learning rules capable of detecting spike patterns even in the presence of jitter noise. In this study, we introduce a simplified suppression-based STDP rule that demonstrates significantly enhanced tolerance to jitter in spike patterns compared to the pair-based STDP rule. Unlike the ideal suppression STDP rule, characterized by an exponential learning window and requiring high-resolution synapses, the simplified rule limits the synaptic efficacy update to a single bit at any given instant. Moreover, it employs 4-bit fixed-point synapses, facilitating straightforward implementation in neuromorphic hardware.

Index Terms—Triplet STDP rule, suppression STDP rule, neuromorphic computing, low-resolution synapse, spike pattern detection, jitter noise, lateral inhibition.

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I. INTRODUCTION

Inspired by the neuronal networks in the brain, the computational backbone of neuromorphic chips comprises electronic versions of neuronal cells and synapses [24], [4]. They attract significant attention as a pivotal technology for significantly enhancing the energy efficiency of artificial intelligence (AI) tasks, addressing an urgent societal concern. Chips that use low-resolution synapse circuits for learning and not just for inference are crucial for Edge AI applications. In this context, the development of high-efficiency learning models emerges as a crucial technological imperative. The most widespread on-chip learning mechanism integrated in neuromorphic chips is spike-timing-dependent plasticity (STDP) [5], [18]. STDP-based learning rules act locally on the synapse, modifying its efficacy primarily based on the spike timings of the pre- and postsynaptic neurons. The most widely implemented STDP rule is the pair-based STDP, which models the modification in synaptic efficacy by considering the most recent pairs of pre- and postsynaptic spikes. Synaptic efficacy is potentiated if a presynaptic spike occurs before a postsynaptic one and is depressed for the reverse order. However, it captures only a limited aspect of synaptic plasticity. In the presence of multiple pre- and postsynaptic spikes, multi-spike STDP rules provide a better fit to the observed data in the brain [25], [26], [27]. Efficacy modification in multi-spike STDP rules is counterintuitive compared to the pair-based STDP rule. For example, in spike triplets of the form pre-post-pre and post-pre-post with equal delays between the spike pairs, the efficacy modification is not equal, as one would expect based on the pair-based STDP rule. Two widely recognized STDP rules that involve multiple spikes are the triplet rule and the suppression rule [25], [8]. The triplet rule was proposed to model

observation in hippocampal slices. Here, in addition to the effects of the pair-based STDP rule, additional depression due to earlier presynaptic spikes and additional potentiation due to earlier postsynaptic spikes are also included. The suppression-based STDP rule was inspired by observations made in rat's cortical slices [8]. These observations markedly differed from the triplet STDP rule. It was noted that the modification in efficacy for each spike pair, following the pair-based STDP rule, was suppressed by the preceding spikes. Similar to the pair-based STDP rule, in the multi-spike STDP rules, the magnitude of efficacy modification falls exponentially with an increasing interval between the spike pairs and the preceding spikes. Stability, competition, and population dynamics of these multi-spike learning rules have been explored in previous studies [3], [22]. However, we are not aware of any studies where these multi-spike rules have shown advantages in terms of information processing over the pair-based STDP rule. In this study, we propose a simplified suppression-based STDP rule with enhanced jitter tolerance for spatiotemporal spike pattern detection using low-resolution synaptic weights. This mechanism addresses the essential requirements of improved learning ability and seamless integration into neuromorphic hardware, critical for chip-scale network scalability.

In [19], it was empirically shown that a single neuron empowered with the pair-based STDP rule can detect repeating spatiotemporal spike patterns hidden at irregular intervals in equally dense distractor spike trains modeled by an inhomogeneous Poisson process (See Fig. 1). These patterns adhere to temporal coding, and such repeating spike patterns with precision on the scale of milliseconds are common in the brain [17]. Additionally, in [20], it was shown that a layer of STDP-empowered neurons, which laterally inhibit each other can effectively classify multiple spike patterns. This network motif of STDP with lateral inhibition, observed in various brain regions, is widely accepted as one of the most prevalent learning motifs in the brain [7], and has been used in many SNN architectures [6]. The spike train models used in these studies also included additional noise and jitter roughly corresponding to synaptic noise. The classification performance of both single and multi-neuron networks was observed to degrade considerably with increased jitter in the timing of the spike within the patterns [19], [20]. As spiking neural networks (SNN) seldom produce temporally identical spike trains in response to the same input, it is crucial to have learning rules that can distinguish target spike patterns from background noise, even when multiple instances of the spike patterns are not precisely alike. In this paper, utilizing the same input spike train models and network configurations as in [19] and [20], we show via numerical simulations that our proposed simplified suppression-based STDP learning rule is highly tolerant to jitter noise in the timing of spike patterns. Furthermore, since STDP circuits are notoriously difficult to scale in mixed-signal neuromorphic chips due to the requirement of synapse circuits with high bit resolution (8 to 10 bits) [2], our proposed learning rule is simplified compared to the ideal suppression-based STDP model and requires low-

resolution synapses. Results are presented using 3-bit and 4-bit fixed-point synapses. Low-resolution synapses are more amenable to scaling of network size in both CMOS as well as novel non-volatile memory technologies such as memristors.

The rest of the manuscript is organized as follows: the Material and Methods section begins with an explanation of the network configurations and the input spike train model. This is followed by a description of the neuron model, pair-based STDP rule, and the simplified suppression-based STDP learning rule. Section III presents simulation results highlighting its capability. The manuscript concludes with a discussion of the results and planned future works.

II. MATERIALS AND METHODS

A. Network configuration

Simulations were performed for two configurations of SNN. The first network comprised a single leaky integrate-and-fire (LIF) neuron receiving spikes from 2048 afferents. The input spike trains contained one 50 ms long hidden spike pattern, repeating at irregular intervals encoded by 1024 afferents (See Fig. 1). This configuration mirrors the network described in [19] and we will call this 1N-1P network. The second network comprised 9 LIF neurons with lateral inhibition between them, each receiving spike trains from 2048 afferents. In this setup, the input spike trains contained three distinct spike patterns, each independently encoded by 1024 afferents (See Fig. 2). This configuration resembles the network described in [20] and is designated as the 9N-3P network. It's noteworthy that the same afferents were not utilized to encode all three spike patterns. The pattern appearance rate for each spike pattern was 10% in the baseline scenario. This means that each spike pattern occupied 1/10th of the total duration of the input spike train, i.e., a 225-s long spike train comprised 450 50-ms long spike patterns. Consequently, the three spike patterns occupied 3/10th of the total simulation time in the 9N-3P network. In both configurations (1N-1P and 9N-3P), the parameters characterizing the properties of the spike trains that control the difficulty of the pattern detection/classification are the standard deviation of the jitter, noise frequency, and the pattern appearance rate.

The 1N-1P network configuration showcases the stand-alone discriminative capability of the proposed simplified suppression-based STDP rule, while the 9N-3P network illustrates its scalability for detecting multiple patterns. The selection of nine neurons and three patterns mirrors the network configuration used in the previous study employing the pair-based STDP rule for the same task [20], allowing for direct comparison. These choices can be adjusted as needed.

B. Input Spike Trains

The input spike trains used in this study were developed following biologically plausible prerequisites. A detailed description of the procedure to generate these spike trains is found in [19] and [20]. They were modeled using an inhomogeneous Poisson process known to roughly capture the statistical properties of spikes in the brain. The spike trains

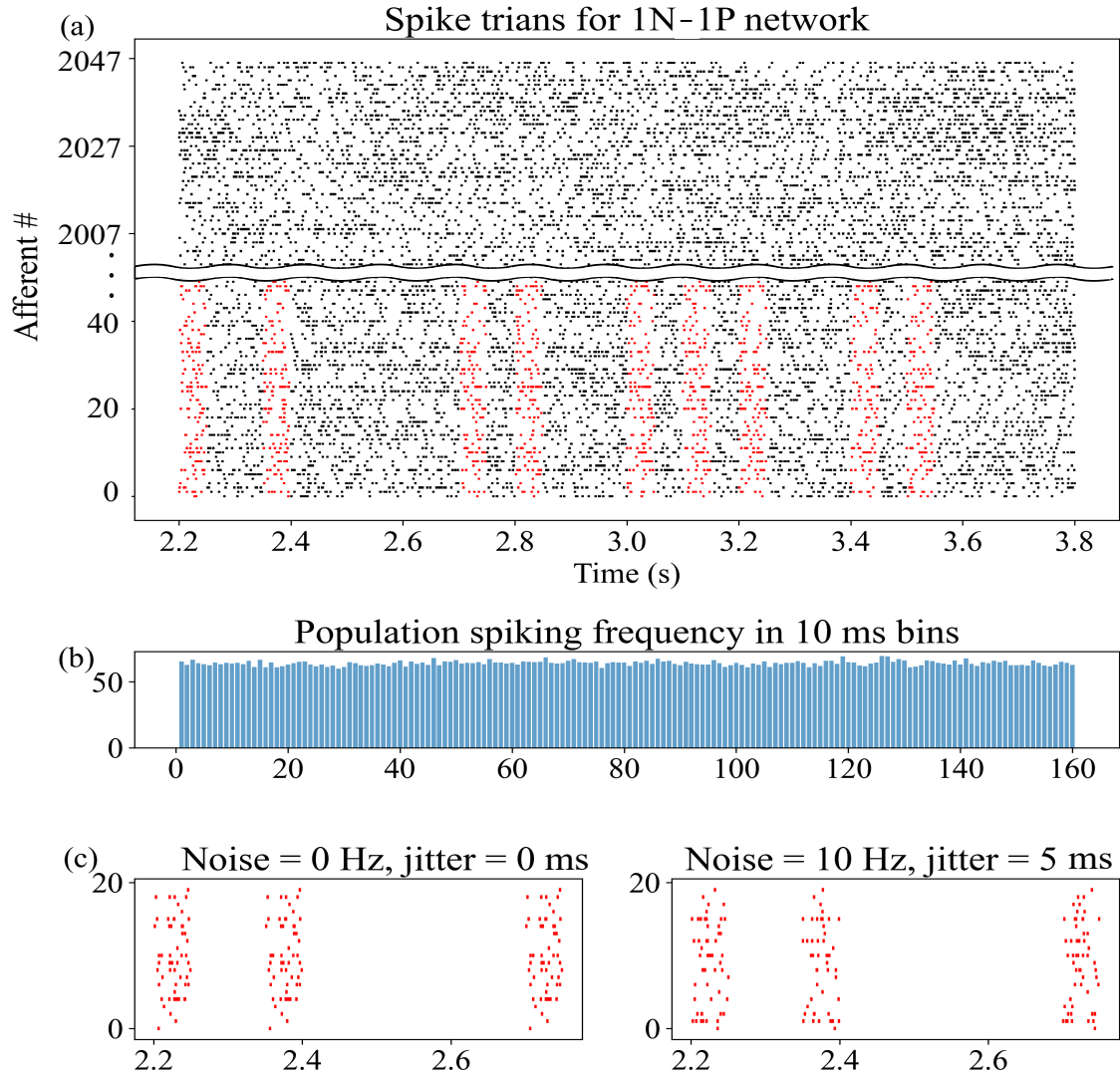


Fig. 1. (a) The raster plot illustrates the input spike trains. The embedded spike patterns encoded by 1024 out of 2048 afferents are highlighted in red. These spike trains exhibit jitter in spike timing within the patterns, with a standard deviation of 1 ms. Each afferent also has an additional 10 Hz noisy spikes. (b) The afferents, on average, maintain a spiking frequency of 64 Hz. These spike patterns are temporally coded, specifically characterized by the spike timing of the afferents. (c) Multiple instances of the patterns with different values of noise and jitter for 20 afferents. With higher values of noise and jitter, multiple instances of the pattern become significantly different from one another making the pattern detection challenging.

were composed of 2048 afferents spiking continuously and independently for 225 s. Arbitrary 50 ms long spike patterns were inserted at irregular intervals in the input spike trains. Only half of the total afferents (1024 out of 2048) were used to encode the spike patterns. In other words, the 2048 afferents spiked stochastically and independently for most of the time. But occasionally, 1024 afferents left the stochastic spiking mode and fired in a precise spiking pattern for a duration of 50 ms. The number of the spike patterns that appear in a 225 s spike train is controlled by a model parameter to control the pattern appearance rate. For example, if the pattern appearance rate is 25%, the spike patterns would occupy a total of 1/4th duration of the 225 s-long spike train. The population spike

rate inside and outside the patterns was roughly the same, and the repeating spike patterns were characterized solely by the timing of the spikes. The average population spike rate was 54 Hz. To make the pattern detection more challenging, an additional 10 Hz Poisson spiking activity was added to each afferent. Furthermore, a jitter with a standard deviation of 1 ms was included in the spike timing of each afferent within each instance of the pattern. With the 10 Hz noise included, the population average spiking rate became 64 Hz. Without the inclusion of the noise and jitter, each instance of the pattern within the spike train would be the same (See Fig. 1(c)). In [19], it was shown that a single LIF neuron with the pair-based STDP rule learns to detect these spike patterns.

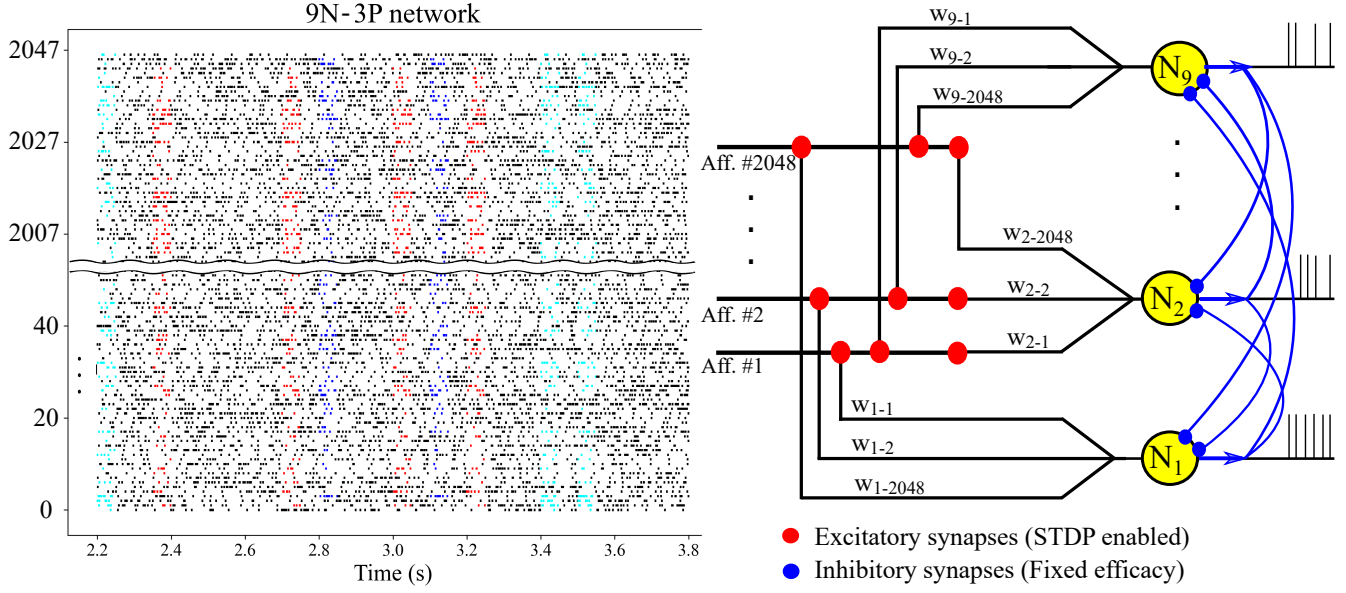


Fig. 2. The raster plot on the left illustrates the input spike trains with three embedded spike patterns distinguished by three different colors. The network on the right comprises nine LIF neurons (N1-N9) that laterally inhibit each other (blue connections). The neurons are simulated by 2048 synapses that receive input spike trains. The excitatory synapses are equipped with STDP learning and have efficacies bounded between 0 and 1. In contrast, the inhibitory synapses do not possess learning capability, and their efficacies remain constant.

In the initial phase of the simulation run, the neuron spikes randomly inside and outside the spike patterns. However, over time due to successive coincidences of repeatedly present spike patterns, the STDP rule potentiates a selective group of synapses associated with the spike patterns and depresses the synaptic efficacy of remaining afferents. Thus, in time, the neuron learns to spike only when it encounters a spike pattern. A greater value of jitter increases dissimilarity among various instances of the spike patterns, thereby raising the challenge of successfully detecting these patterns. The difference in temporal structure in multiple instances of the spike patterns with and without noise and jitter is shown in Fig. 1(c). The pattern appearance rate also controls the difficulty of pattern detection. A lower rate makes it more challenging for the network to learn the spike patterns, as the neuron is exposed to fewer instances of spike patterns, primarily encountering noisy spikes. In the result section, we present simulation results to quantify the effect of jitter, noise, and pattern appearance rate on the network's capacity to recognize spike patterns. Like the reference studies [19] and [20], the 225 s long input spike trains were repeated twice to make it 450 s long for the 1N-1P network and thrice to make it 675 s long for the 9N-3P network. As the period of 225 s is significantly longer than 50 ms, this repetition is not expected to have any significant effect on the learning process.

C. Neuron Model

Following reference studies [19] and [20], the SRM formulation of the LIF neuron model was employed for simulation [14]. The excitatory postsynaptic potential (EPSP) due to a presynaptic spike at time t_j is given by

$$\varepsilon(t - t_j) = K \cdot \left(\exp \left(-\frac{t - t_j}{\tau_m} \right) - \exp \left(-\frac{t - t_j}{\tau_s} \right) \right) \cdot u(t - t_j), \quad (1)$$

where $\tau_m = 10$ ms and $\tau_s = 2.5$ ms are the membrane and synapse time constants, respectively. The function u is the Heaviside step function. An arbitrary constant K sets the maximum amplitude of the EPSP kernel. In this study, K was selected to set the maximum amplitude to 1. Upon reaching a preset threshold (T), the neuron generates a spike. After each postsynaptic spike at time t_i , the spike profile and after-hyperpolarization are defined as follows:

$$\eta(t - t_i) = T \left(K_1 \exp \left(-\frac{t - t_i}{\tau_m} \right) - K_2 \left\{ \exp \left(-\frac{t - t_i}{\tau_m} \right) - \exp \left(-\frac{t - t_i}{\tau_s} \right) \right\} \right) u(t - t_i), \quad (2)$$

where constants $K_1=2$ and $K_2=4$ set the amplitude of the spike and the after-hyperpolarization, respectively. In the 9N-3P network, the spiking neuron inhibits every other neuron by adding an inhibitory postsynaptic potential (IPSP) to their membrane potential. The IPSP for each postsynaptic spike at time t_k is given by

$$\mu(t - t_k) = -\alpha \cdot T \cdot (\varepsilon(t - t_k)) \quad (3)$$

The EPSP kernel with a scaling factor α is used to model the IPSP kernel. Combining the three kernels, the membrane potential of the neuron at any time instant is given by

$$v = \eta(t - t_i) + \sum_{j/t_j > t_i} w_j \cdot \varepsilon(t - t_j) + \sum_{k/t_k > t_i} \mu(t - t_k). \quad (4)$$

Here w_j represents the synaptic efficacies of the excitatory synapses, randomly initialized and constrained between 0 and 1. All three kernels are truncated to zero when $t - t_{i,j,k}$ exceeds $7\tau_m$. Additionally, after each postsynaptic spike, the EPSP and IPSP kernel of spiking neurons are reset. The inhibitory synapses do not have learning capability and their efficacy remains constant.

D. STDP Rules

1) *Pair-based STDP rule*: The pair-based STDP rule modifies synaptic efficacy according to the causality of pre- and postsynaptic spikes pairs. A causal pre-post pair causes potentiation and an acausal post-pre pair causes depression. The modification exhibits an exponential dependence on the spike interval, as depicted in Fig. 3(a). This is mathematically expressed as:

$$\Delta w_j = \begin{cases} a^+ \cdot \exp\left(\frac{\Delta t}{\tau^+}\right) & \text{if } \Delta t \geq 0 \quad (\text{LTP}), \\ a^- \cdot \exp\left(-\frac{\Delta t}{\tau^-}\right) & \text{if } \Delta t < 0 \quad (\text{LTD}), \end{cases} \quad (5)$$

where a^- and a^+ are the learning rates controlling depression and potentiation, respectively.

The temporal spread of the learning window is controlled by the time constants τ^- and (τ^+) , and δ_t is the time interval between pre- and postsynaptic spikes. It is positive for causal pre-post pairs and negative for acausal post-pre pairs. The values of these parameters were fixed at $a^+ = 0.03125$, $a^- = 0.85 \cdot a^+$, $\tau^+ = 16.8$ ms, $\tau^- = 33.7$ ms. These values keep the function biased towards depression as $a^+ \tau^+ < a^- \tau^-$. To implement this rule in CMOS-only mixed-signal neuromorphic chips, the exponential learning window is quantized into multiple levels as shown by the blue trace in Fig. 3(a).

2) Suppression-based STDP rule:

a) *Ideal Model*: In synaptic plasticity experiments performed on cortical slices, it was observed that for a pre-post spike triplet, the first pre-post spike pair led to potentiation and the next post-pre spike pair led to depression. This is in line with the pair-based STDP rule. However, the magnitude of depression was influenced by the first presynaptic spike. The closer the interval between the first presynaptic spike to the post-pre spike pair, the smaller the magnitude of depression induced. In other words, the preceding presynaptic spike suppresses the effect of the successive post-pre spike pair. Similar suppression effects were observed for a post-pre-post triplet, where the potentiation of pre-post spike pair was influenced by the preceding postsynaptic spike. This is mathematically expressed as

$$\Delta w = (1 - \exp(-\frac{\Delta t_{\text{pre}}}{\tau_{\text{pre}}})) \cdot (1 - \exp(-\frac{\Delta t_{\text{post}}}{\tau_{\text{post}}})) \cdot \begin{cases} a^+ \cdot \exp(-\frac{\Delta t}{\tau^+}) & \text{if } \Delta t \geq 0 \quad (\text{LTP}), \\ a^- \cdot \exp(\frac{\Delta t}{\tau^-}) & \text{if } \Delta t < 0 \quad (\text{LTD}), \end{cases} \quad (6)$$

where Δt_{pre} denotes the time gap between the presynaptic spike within the pair and its previous presynaptic spike and Δt_{post} denotes the time gap between the postsynaptic spike within the pair and its previous postsynaptic spike. The time constants τ_{pre} and τ_{post} determine the reduction in the effect of suppression with increasing values of time intervals Δt_{pre} and Δt_{post} , respectively. Other parameters are the same as in the pair-based STDP rule.

b) *Simplified Rule*: We simplify the learning rule described above by replacing the exponential windows in the learning function with rectangular ones. This restricts the efficacy update at any point in time to a single bit and the learning rule simplifies to the following:

$$\Delta w = \begin{cases} +1 \text{ bit} & \text{if } \Delta t \geq 0 \text{ and } \Delta t < t_{\text{pre}} \\ & \text{and } \Delta t_{\text{post}} > St_{\text{post}} \quad (\text{LTP}), \\ -1 \text{ bit} & \text{if } \Delta t < 0 \text{ and } \Delta t < t_{\text{post}} \\ & \text{and } \Delta t_{\text{pre}} > St_{\text{pre}} \quad (\text{LTD}), \end{cases} \quad (7)$$

where for a spike pair, t_{pre} (t_{post}) is the maximum delay of the postsynaptic (presynaptic) spike after a presynaptic (postsynaptic) spike within which potentiation (depression) is allowed to occur. St_{post} (St_{pre}) denotes the maximum delay of the postsynaptic (presynaptic) spike in any pre-post (post-pre) spike pair after a preceding postsynaptic (presynaptic) spike within which potentiation (depression) due to the post-pre (pre-post) spike pair is suppressed. In other words, the exponential windows controlled by the parameters τ^+ , τ^- , τ_{pre} and τ_{post} are replaced by rectangular windows with parameters t_{pre} , t_{post} , St_{pre} , and St_{post} , and in this study, their values were set at 5.7 ms, 28.5 ms, 17.1 ms, and 20.1 ms, respectively. This is pictorially represented in Fig. 3(b).

III. RESULTS

Simulations were performed using both the pair-based STDP rule with 64-bit floating point resolution synapses and the simplified suppression-based STDP rule with 3- and 4-bit fixed point synapses. Firstly, the simulation results of the 1N-1P network are presented, followed by the 9N-3P network.

1) *1N-1P network*: The three model parameters, standard deviation of the jitter, noise, and the pattern appearance rate configured the characteristics of the generated spike trains. We chose spike trains with 10 Hz noise, jitter with a standard deviation of 1 ms, and a 10% pattern appearance rate as the baseline. From this baseline, we generated spike trains by varying one of the three parameters. This was done to quantify the effect of each parameter separately. The parameter

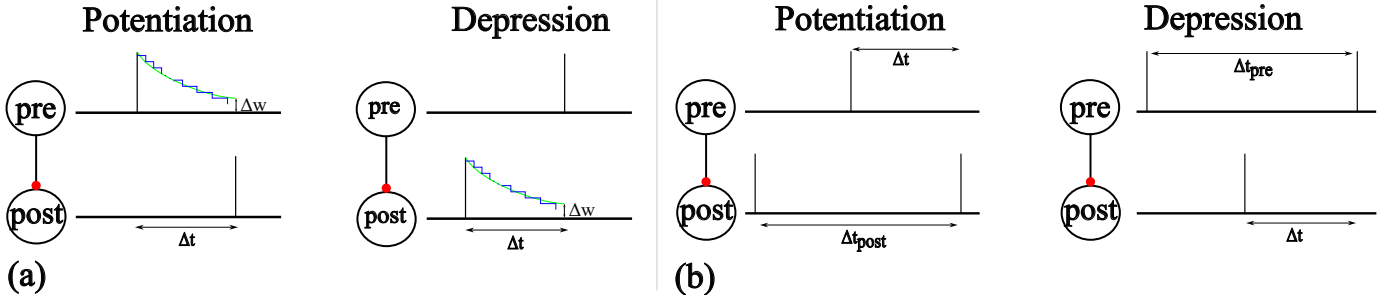


Fig. 3. (a) Efficacy modification in the pair-based STDP rule. Potentiation occurs for causal pre-post spike pairs and depression for acausal post-pre pairs. The modification in efficacy has an exponential dependence interval between the spike pairs. (b) A representation of the simplified suppression-based STDP rule. Potentiation occurs if $\Delta t > t_{pre}$ and $\Delta t_{post} > St_{post}$, and depression occurs if $\Delta t > t_{post}$ and $\Delta t_{pre} > St_{pre}$. For causal spike pairs Δt is positive and for acausal pairs it is negative. The efficacy modification is always ± 1 bit.

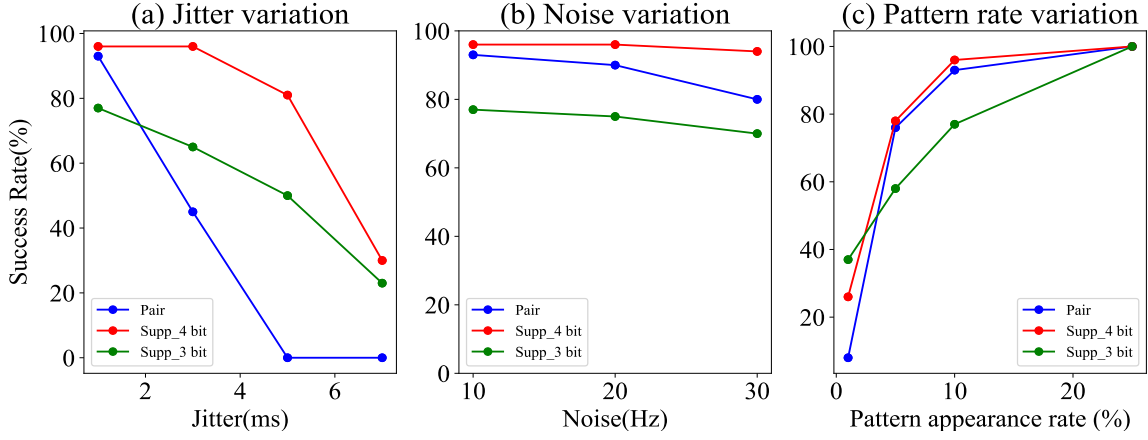


Fig. 4. Performance of 1N-1P network with pair-based STDP rule using 64-bit floating point synapse resolution (blue data points) and simplified suppression-based STDP rules with the latter using 4-bit (red data points) and 3-bit (green data points) fixed-point synapses. The success rates obtained for input spike trains with (a) jitter with a standard deviation of 1 ms, 3 ms, 5 ms, and 7 ms (b) noise of 10 Hz, 20 Hz, and 30 Hz (c) pattern appearance rate of 25%, 10%, 5%, and 1%. 100 simulation runs were performed for each data point in the figure. All runs utilized the same learning parameters for the corresponding STDP rules.

values used were jitter with a standard deviation of 1 ms, 3 ms, 5 ms, and 7 ms, noise of 10 Hz, 20 Hz, and 30 Hz, and pattern appearance rate of 25%, 10%, 5%, and 1%. For each configuration, 100 simulation runs were performed with 100 different inputs generated by the procedure in Sec. II (B). The performance was evaluated based on the hit rate (neuron spikes within the spike pattern) and false alarms (neuron spikes outside the spike pattern) in the last 225 seconds of each run. The success criterion was chosen to be a hit rate greater than 98% hit rate and the presence of zero false alarms like the reference study [19]. The results are summarized in Fig. 4.

For the configurations in which the noise and pattern appearance rates are varied, the performance of all the learning rules is similar, with the simplified suppression-based STDP rule performing slightly better. However, with higher jitter noise in the spike patterns, the suppression-based STDP rule performs significantly better than the pair-based STDP rule. As higher resolution synapses perform better in pattern detection tasks, consequently, the suppression-based STDP rule exhibits poorer performance with 3-bit synapses compared to 4-bit synapses. However, it still outperforms the pair-wise STDP rule with 64-

bit synapse resolution at higher jitter values. The parameters of the STDP rules were kept the same across all configurations. In configurations where the noise was increased from the baseline value of 10 Hz, only the spiking threshold of the LIF neuron was increased. This is natural because the average population spike rate of the input increases with higher noise levels and the threshold must be appropriately adjusted. To determine suitable threshold values for spike trains with increased noise, we conducted simulations using varying threshold values. These thresholds were incrementally adjusted from the baseline value in steps until optimal performance was attained in a few simulation runs. All other network parameters were similarly tuned manually. As shown in Fig. 4, the suppression-based learning rule also performs reasonably well with 3-bit synapses, but the learning parameters used with 4-bit synapses did not perform well with 3-bit synapses. The parameters t_{pre} , t_{post} , St_{pre} , and St_{post} were set at 7.7 ms, 40.5 ms, 24 ms, and 20.1 ms, respectively for the three bit case.

2) *9N-3P network*: The lateral inhibitory connections in the 9N-3P network direct the neurons to learn different patterns or different sections of the same 50 ms-long spike pattern

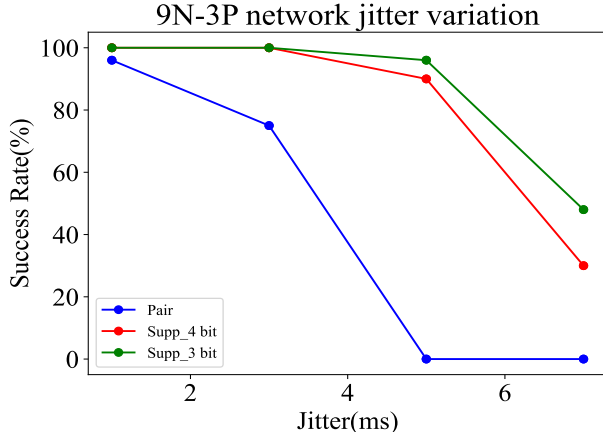


Fig. 5. Performance of 9N-3P network with pair-based STDP rule using 64-bit floating point synapse resolution (blue data points) and simplified suppression-based STDP rules with the latter using 4-bit (red data points) and 3-bit (green data points) fixed-point synapses. The success rates obtained for input spike trains with jitter with a standard deviation of 1 ms, 3 ms, 5 ms, and 7 ms.

[20]. The performance trends observed in the 1N-1P network extend to the 9N-3P network. Like the reference study [20], the success criterion in the 9N-3P network was chosen to be a hit rate greater than 98% and false alarms under 1 Hz in the last 225 s of the run. A hit rate of 100% implies that all the instances of the three spike patterns are detected by at least one of the nine neurons in the last 225 s of the simulation. In most cases, multiple neurons learn different sections of the same 50-ms long spike pattern. In the baseline condition with 10 Hz noise, jitter with a standard deviation of 1 ms, and 10% pattern appearance rate for each pattern, all rules have similar performance. However, with a higher jitter value of 5 ms, the simplified suppression-based rule significantly outperforms the pair-based STDP rule as shown in Fig. 5. The 9N-3P network demonstrates superior overall performance compared to the 1N-1P network. In the 9N-3P network, an average of three neurons can learn one pattern, whereas the 1N-1P network only allows a single neuron to learn a pattern. This distinction is evident in the results obtained with the suppression-based STDP rules, where the performance remains similar for both 3-bit and 4-bit synapses.

IV. DISCUSSION AND CONCLUSION

Our results empirically show that the simplified suppression-based STDP rule with 3 and 4-bit fixed-point synapses is significantly more tolerant to jitter in spike patterns than the pair-based STDP rule with 64-bit floating point resolution synapses available in digital computers. The results from the 9N-3P network demonstrate that the proposed rule is scalable to a network of multiple neurons needed to classify an arbitrary number of patterns and perform classification tasks on larger neuromorphic datasets, utilizing an architecture similar to that described in [6]. The proposed learning rule satisfies both the criteria of superior learning capability and ease of

implementation in neuromorphic hardware, which are essential for scaling networks on fabricated chips.

The superior performance of the simplified suppression-based STDP rule even in the presence of high jitter noise can be intuitively understood due to slow and selective potentiation of synapses associated with spike pattern. The chosen parameters bias both STDP rules towards depression. In other words, in the absence of repeating spike patterns the synaptic efficacy for all synapses would depress in time. With repeated coincidences of spike patterns, STDP-based rules potentiate a group of synapses that are active in temporal proximity within the spike pattern. This corresponds to a section of the 50 ms-long spike pattern and the neuron then only spikes when this group of synapses activates in proximity in repeating instances of the spike pattern. With increasing jitter, the temporal structure of spikes within the pattern varies significantly across multiple instances of the spike patterns (See Fig. 1(c)). In the pair-based STDP rule, as every spike pair leads to a modification in the efficacy, a potentiation of a group of synapses associated with the pattern is highly likely to be reversed by a successive depression than be reinforced by the next instance of the spike pattern because of its temporal distortion. Whereas in the suppression-based STDP rule, the modification in efficacy does not occur for every spike pair. The suppression effect of earlier spikes acts as a filter to regulate the depression of synapses that have been potentiated and vice versa. Thus, in time a larger number of synapses associated with a section of the pattern get a chance to be potentiated and the neuron learns to detect the spike patterns. In the spike train configurations with lower pattern appearance rate or higher noise, as the temporal structure of the spikes within the pattern does not change significantly in multiple instances of the spike pattern, similar performance is observed with both rules. As the network parameters were tuned manually, we tried various possible sets of parameter values for all three rules, and chose ones that gave reasonably good performance across all configurations of the input spike trains. However, the performance of the network with high jitter noise could not be improved for any possible set of parameter values when using the pair-based STDP rule. In this study we presented empirical results, in the future, we will analytically explore the dynamics of this learning rule at a population level using similar spike train models with temporally coded spike patterns.

It is generally known that the performance of pair-based STDP rules in similar pattern detection tasks deteriorates with synapses of lower resolution [20], [16]. Developing learning rules that maintain performance with low-resolution synapses is crucial because, in mixed-signal neuromorphic chips that use digital-to-analog converters (DACs) and digital memory to implement and store synaptic efficacy, the size of the synapse circuit doubles for every single bit increase in the synapse resolution [24], [21], [9], [11]. Novel non-volatile memory devices such as memristors and ferroelectric field-effect transistors (FeFET) are believed to be a potential solution to implement compact high-resolution synapses [23],

[28]. However, these devices are still under research, they also do not have stable resolution greater than 4 to 5 bits, and they come with hardware implementation overheads. As the proposed learning rule uses only 4-bit fixed point synapses, it is more amenable to scaling in mixed-signal neuromorphic hardware. Furthermore, the simplified suppression-based STDP rule restricts the update in synaptic efficacy at any time instant to 1 bit. Thus, in its circuit implementation, the efficacy update can be done with a 4-bit up-down counter. In contrast, the ideal STDP rules due to their quantized step-like learning function require additional circuit elements such as adder, subtractor, and look-up tables, even if the resolution of synapses is 4-bit. This increases the circuit's complexity, area, and energy consumption. A circuit for implementing the pair-based STDP rule, featuring a rectangular learning window rather than an exponential one and utilizing 4-bit synapse resolution, is described in [12]. The simplified suppression-based STDP rule can be implemented by appending this circuit with additional components to measure Δt_{pre} and Δt_{post} .

A limitation of this study is that learning parameters were manually tuned, with different optimal settings for the 3-bit and 4-bit suppression STDP rules. Future work will explore algorithms, including evolutionary approaches like those in [15], to identify the most suitable network parameters for similar tasks with spatiotemporal spike trains, such as DVS Gesture [1].

The STDP phenomenon was first observed over two decades ago [5], [18]. In addition to higher-order spike interactions, various other factors that influence STDP learning have been discovered. Some of these include the presence of neurotransmitters like dopamine and acetylcholine, instantaneous post-synaptic membrane potential, and the location of synapses on the dendritic tree. An adaptive STDP rule inspired by the influence of neurotransmitters on the learning window is presented in [10], [13] and its neuromorphic implementation is described in [12]. There is no single learning rule that can account for all these factors but given their presence in the brain, modeling these additional factors for implementation in neuromorphic chips will not only make them more biologically plausible but is also likely to improve their discriminatory power. This study was restricted to simulation results, in the future, this learning rule will be implemented in a mixed-signal neuromorphic chip.

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